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Evaluating the Gap Between Test-Passing and Semantically Correct Patches in Large Language Model-Based Automated Program Repair

1. Research Gap Statement

Paragraph 1: Background and State of the Art

Automated Program Repair (APR) aims to automatically identify and correct software defects, reducing the time and effort required for manual debugging. The emergence of Large Language Models (LLMs) has expanded APR capabilities by enabling models to interpret bug reports, analyze source code, generate candidate patches, and interact with software testing tools. Zhang et al. (2024), in *A Systematic Literature Review on Large Language Models for Automated Program Repair*, reviewed research on LLM-based APR and identified its major techniques, repair scenarios, and remaining challenges. Existing APR workflows commonly involve fault localization, patch generation, patch validation, and patch correctness assessment. Passing available test cases is an important initial indicator that a generated patch may be useful. However, test execution alone cannot establish that the patch correctly implements the intended behavior in every relevant situation. This distinction makes the reliability of patch evaluation an important issue for the practical adoption of LLM-based APR.

Paragraph 2: The Specific Unresolved Gap

A specific research problem is the potential discrepancy between a patch that passes the available test suite and a patch that is semantically correct. A test suite may cover common execution paths without adequately representing boundary conditions, unusual inputs, or interactions with other program components. Consequently, a generated patch may satisfy the existing tests while leaving the original defect partially unresolved or introducing unintended behavior. This creates an evaluation problem: if a system is judged primarily by test-passing results, its apparent repair performance may be higher than its actual semantic correctness. Although APR research recognizes patch validation and correctness as

important stages, the extent of this discrepancy may vary across bug types, test-suite coverage levels, and repair methods. A controlled empirical evaluation is therefore needed to quantify the relationship between test-passing outcomes and independently assessed patch correctness.

Paragraph 3: Why Existing Approaches May Be Insufficient

Test-based validation is practical because automated test suites provide repeatable and relatively inexpensive checks. Nevertheless, a passing test suite can only establish that the patch satisfies the assertions exercised by those tests. It does not necessarily demonstrate that the implementation conforms to the complete functional specification. Additional validation techniques, such as regression testing, mutation testing, differential testing, and manual semantic assessment, can provide complementary evidence, but they introduce different implementation costs and evaluation requirements. Furthermore, comparing results from different APR systems can be difficult when their benchmarks, test suites, and correctness criteria differ. The LLM4APR literature and existing APR surveys emphasize the importance of understanding evaluation methods and patch correctness, while also identifying challenges in conducting comparable empirical assessments. The remaining empirical question is how strongly the measured performance of LLM-based APR depends on the validation procedure and how much additional correctness evidence is obtained from complementary validation methods.

Paragraph 4: Proposed Intervention and Expected Contribution

This study proposes a controlled evaluation of generated patches using multiple validation criteria. A selected set of software bugs will be repaired using a fixed LLM-based APR configuration. Each generated patch will first be evaluated against the available test suite and then assessed using additional validation evidence, such as regression tests, mutation-based checks, and manual review of functional behavior. The study will distinguish patches that fail existing tests, patches that pass existing tests but fail additional validation, and patches that satisfy the defined correctness criteria. The independent variables will include validation strategy and test-suite coverage condition, while the dependent variables will include the test-passing rate, the proportion of patches that fail additional validation, and the independently assessed correct-patch rate. The expected contribution is a reproducible evaluation protocol that clarifies the limitations of test-passing results as a standalone indicator of repair quality. The study will not assume that any particular validation method guarantees complete correctness; instead, it will measure the additional evidence each method provides within the selected benchmark and experimental conditions.

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2. Research Questions

RQ1: Discrepancy Between Test-Passing and Correct-Patch Rates

Research Question: How large is the discrepancy between the test-passing rate and the independently assessed correct-patch rate of LLM-based APR systems?

- **Independent Variable (IV):** Evaluation criterion, comparing test-suite results with independently assessed patch correctness.
- **Dependent Variables (DVs):**
    - Test-passing rate.
    - Independently assessed correct-patch rate.
    - Percentage-point difference between the two rates.
- **Measurement Method:** Evaluate generated patches using the available automated test suite and a predefined correctness-assessment procedure. Calculate the difference between the test-passing rate and the independently assessed correct-patch rate.

RQ2: Effect of Test-Suite Coverage

Research Question: To what extent does test-suite coverage affect the discrepancy between test-passing results and semantic patch correctness?

- **Independent Variable (IV):** Test-suite coverage condition, comparing the original test suite with an expanded test suite.
- **Dependent Variable (DV):** False-positive patch rate, defined as the proportion of patches that pass the original test suite but fail the additional validation criteria.
- **Measurement Method:** Evaluate the same generated patches under different test-suite conditions. Compare the false-positive patch rates and document the coverage conditions used in each experiment.

RQ3: Effect of Complementary Validation Methods

Research Question: How does the addition of complementary validation methods affect the detection of incorrect patches and the cost of patch evaluation?

- **Independent Variable (IV):** Validation strategy, including the original test suite alone, additional regression testing, mutation testing, or a feasible combination of validation methods.
- **Dependent Variables (DVs):**
    - Number and proportion of incorrect patches detected.
    - Proportion of patches satisfying the predefined correctness criteria.
    - Evaluation time per patch.
- **Measurement Method:** Apply each feasible validation strategy to the same benchmark patches under consistent evaluation conditions. Compare detection outcomes and evaluation costs across strategies.

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3. Feasibility, Novelty, and Impact

3.1 Feasibility

The study is potentially feasible because existing APR benchmarks and automated testing tools can support the evaluation of generated patches. A fixed LLM-based APR configuration and a clearly defined set of benchmark instances can help control experimental variation. However, the final scope must account for the availability of additional test suites, the effort required for independent correctness assessment, and the computational resources needed for patch generation and validation.

3.2 Novelty

The potential novelty lies in conducting a controlled empirical evaluation of the discrepancy between test-passing outcomes, additional validation results, and explicitly defined patch-correctness criteria. The study should not claim to be the first to investigate this problem without a thorough review of closely related APR research. Its contribution must be distinguished from existing studies through the selected benchmark, experimental conditions, validation strategy, or analysis.

3.3 Impact

The findings could help researchers interpret the evaluation results of LLM-based APR systems more carefully. By examining the relationship between test-passing outcomes and additional correctness evidence, the study may help identify limitations in relying exclusively on existing test suites to evaluate repair quality. The results could also inform the design of more transparent and reproducible patch-evaluation procedures.

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4. References

1. Q. Zhang, C. Fang, Y. Xie, Y. Ma, W. Sun, Y. Yang, and Z. Chen, “A Systematic Literature Review on Large Language Models for Automated Program Repair,” 2024. Available: https://arxiv.org/abs/2405.01466
2. Q. Zhang, C. Fang, Y. Ma, W. Sun, and Z. Chen, “A Survey of Learning-based Automated Program Repair,” 2023. Available: https://arxiv.org/abs/2301.03270